University of Florida, Department of Mathematics

Topology First-Year Exam, January 2019, Part 1

Work all problems.

Theorems and Proofs.

In this section state and prove each theorem. The proofs should be complete without being tedious.

1. Show that a compact Hausdorff space X is normal.

2. State and prove the Contraction Mapping Theorem.

3. Give a proof that if X is any set, there is no function $f : X \to \mathscr{P}(X)$ which is onto. Here $\mathscr{P}(X)$ is the power set of X or the set of all subsets of X.

4. Suppose that X is a Hausdorff space. Show that if $A \subset X$ is compact, then it is closed.

5. Suppose that X and Y are metric spaces and that X is compact. Show that if $f : X \to Y$ is continuous, then f is uniformly continuous.

Examples and Minor Proofs.

In this section, give brief examples, counterexamples, or quick proofs.

6. State Urysohn's Lemma..

7. Is it possible for there to be a continuous function $f : [0, 1] \to [0, 1)$ which is onto?

8. State the Bolzano–Weierstrass Theorem.

9. Define a continuous function $f : C \to [0, 1]$ which is onto where C is the Cantor set and $[0, 1]$ is the unit interval.

10. Give an example of a space X which is T_1 but not Hausdorff.

11. State the Tietze Extension Theorem.

12. Define a quotient map.

13. State the Baire Category Theorem.

14. Let $\{X_\alpha\}_{\alpha\in A}$ be a collection of topological spaces. What is the product topology on $\prod_{\alpha\in A} X_\alpha$?

15. What does it mean for a space X to be first countable? Show that every metric space is first-countable.